

Multiple Choice (10 marks)

1. What pressure is required to compress 5 liters of gas at 1 atm. pressure to 1 liter at a constant temperature?
 - (a) 3 atm
 - (b) 9 atm
 - (c) 5 atm
 - (d) 6 atm
 - (e) 10 atm
2. Calculate the percentage difference between e^x and $1 + x$ for $x = 0.0050$
 - (a) $7.50 \times 10^{-3}\%$
 - (b) $2.50 \times 10^{-3}\%$
 - (c) $0.75 \times 10^{-3}\%$
 - (d) $2.00 \times 10^{-3}\%$
 - (e) $1.25 \times 10^{-3}\%$
3. A stock solution of 12.0 M sulfuric acid is made available. What is the best procedure to make up 100. mL of 4.0 M sulfuric acid using the stock solution and water prior to mixing?
 - (a) Add 67.7 mL of 12.0 M acid to the flask, and then dilute it with 33.3 mL of water.
 - (b) Add 33.3 mL of water to the flask, and then add 66.7 mL of 12.0 M acid.
 - (c) Add 40 mL of water to the flask, and then add 60 mL of 12.0 M acid.
 - (d) Add 60 mL of 12.0 M acid to the flask, and then dilute it with 40 mL of water.
 - (e) Add 50 mL of 12.0 M acid to the flask, and then dilute it with 50 mL of water.
4. Assuming complete ionization, calculate (a) the pH of 0.0001 NHCl, .(b) the pOH of 0.0001 N KOH.
 - (a) pH is 10, pOH is 4
 - (b) pH is 4, pOH is 10
 - (c) pH is 3, pOH is 3
 - (d) pH is 4, pOH is 4
 - (e) pH is 1, pOH is 1
5. Calculate the energy of a hydrogen atom in its ground state ($n = 1$) using the mass of the electron rather than the reduced mass of the hydrogen atom.

- (a) - 13.61 eV
- (b) $1.6022 \times 10^{-19}C$
- (b) - $2.1802 \times 10^{-18}J$
- (b) $9.1095 \times 10^{-31}kg$

True / False (10 marks)

Answer True or False

6. True or False: The complete reaction of VO and Fe_2O_3 can produce more than 5 g of V_2O_5 , exceeding the initial mass of reactants.
7. True or False: The atomic weight of Calcium is approximately 40.2 g/mol based on the charge and mass data provided.
8. True or False: An ester can be formed by the reaction of an amine with an acid, resulting in a different type of organic compound.
9. True or False: The solutions to the equation $x^2 - 7x + 10 = 0$ are $x = 3$ and $x = 4$.
10. True or False: London dispersion forces are primarily due to the permanent dipoles generated by the molecular shapes of covalent compounds.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Calculate the empirical formula of a compound with 51.9% carbon, 4.86% hydrogen, and 43.2% bromine. Discuss the steps taken and the significance of this formula in chemical analysis.
12. Calculate the total volume of liquid water lost by an average adult over 60 years through urine, sweat, fecal matter, and expired air, and discuss the implications of these losses on hydration needs.

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